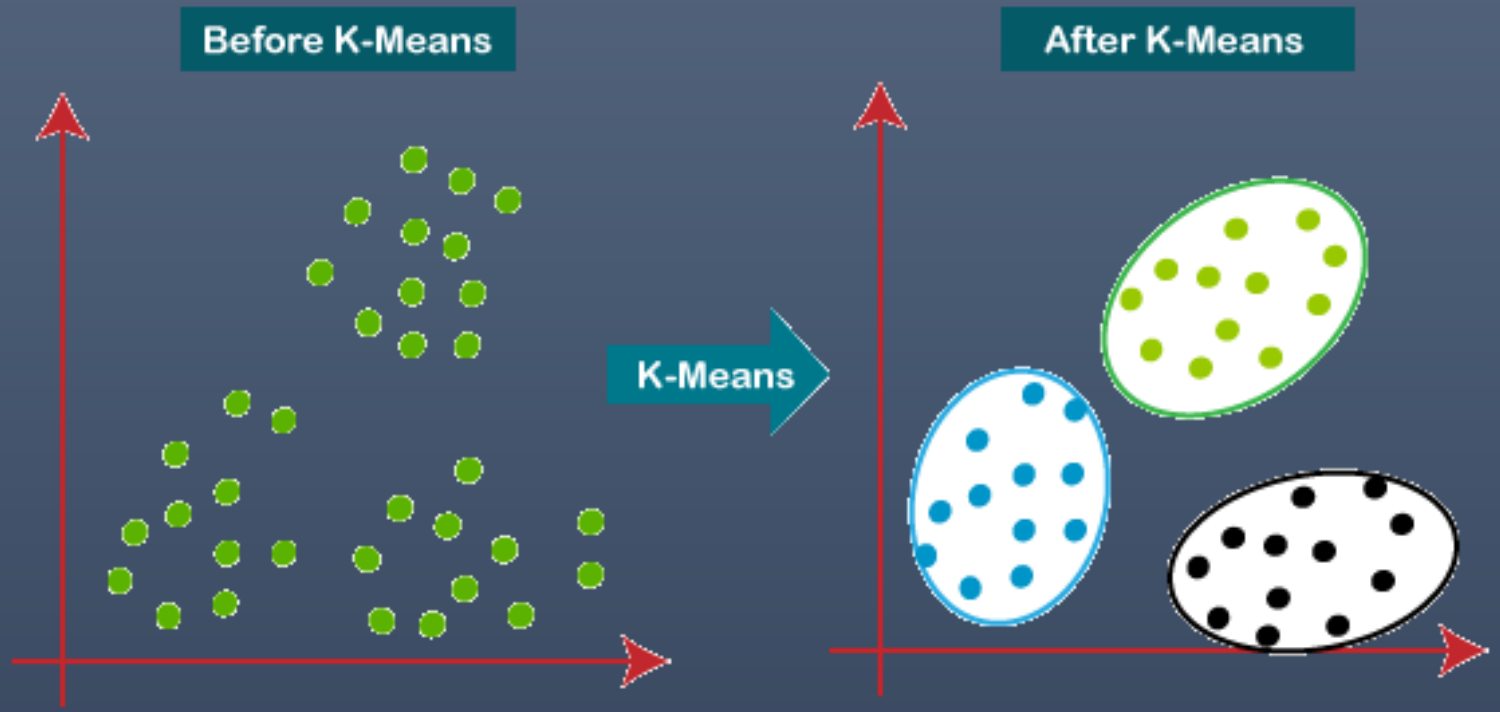
K-Means Clustering:

It aims to partition the data into K clusters, where K is a user-defined parameter. The algorithm iteratively assigns each data point to the nearest cluster centroid and then recalculates the centroids based on the new assignments. It continues this process until the centroids stabilize.



1) Decide clusters
2) Initialize centroids
3) Assign Cluster
4) Re-Calculate Centroid
5) Move centroids
6) Repeat the above until you get correct results
7) Finish

The Euclidean distance between two points (x_1, y_1) and (x_2, y_2) is given by:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Decide Number of Clusters (K):

- Determine the number of clusters KKK you want to create. This is often based on domain knowledge or methods such as the Elbow Method or Silhouette Score to find an optimal number.

Initialize Centroids:

- Randomly select KKK data points from the dataset as the initial centroids (or use other strategies like K-means++ to choose initial centroids more effectively).

Assign Clusters:

- For each data point in the dataset, assign it to the nearest centroid. This forms KKK clusters based on the distance from each data point to the centroids.

Calculate New Centroids:

- Compute the new centroid of each cluster. This is typically done by calculating the mean of all data points assigned to that cluster.

Move Centroids:

- Update the centroid positions to the newly computed centroids from the previous step.

Check Convergence:

- Repeat steps 3 to 5 until the centroids no longer move significantly or a maximum number of iterations is reached. This indicates that the clusters have stabilized and the algorithm has converged.

Finish:

- The algorithm ends when the centroids do not change significantly between iterations or after a predefined number of iterations. The final clusters and centroids are the result of the K-means clustering process.

These steps are repeated iteratively to minimize the within-cluster variance, ensuring that the data points in each cluster are as similar as possible to each other.

Student	Math	Science
St1	24	25
St2	29	20
St3	27	16
St4	69	68
St5	73	61
St6	62	70
St7	82	93
St8	83	84
St9	88	82
St10	99	84

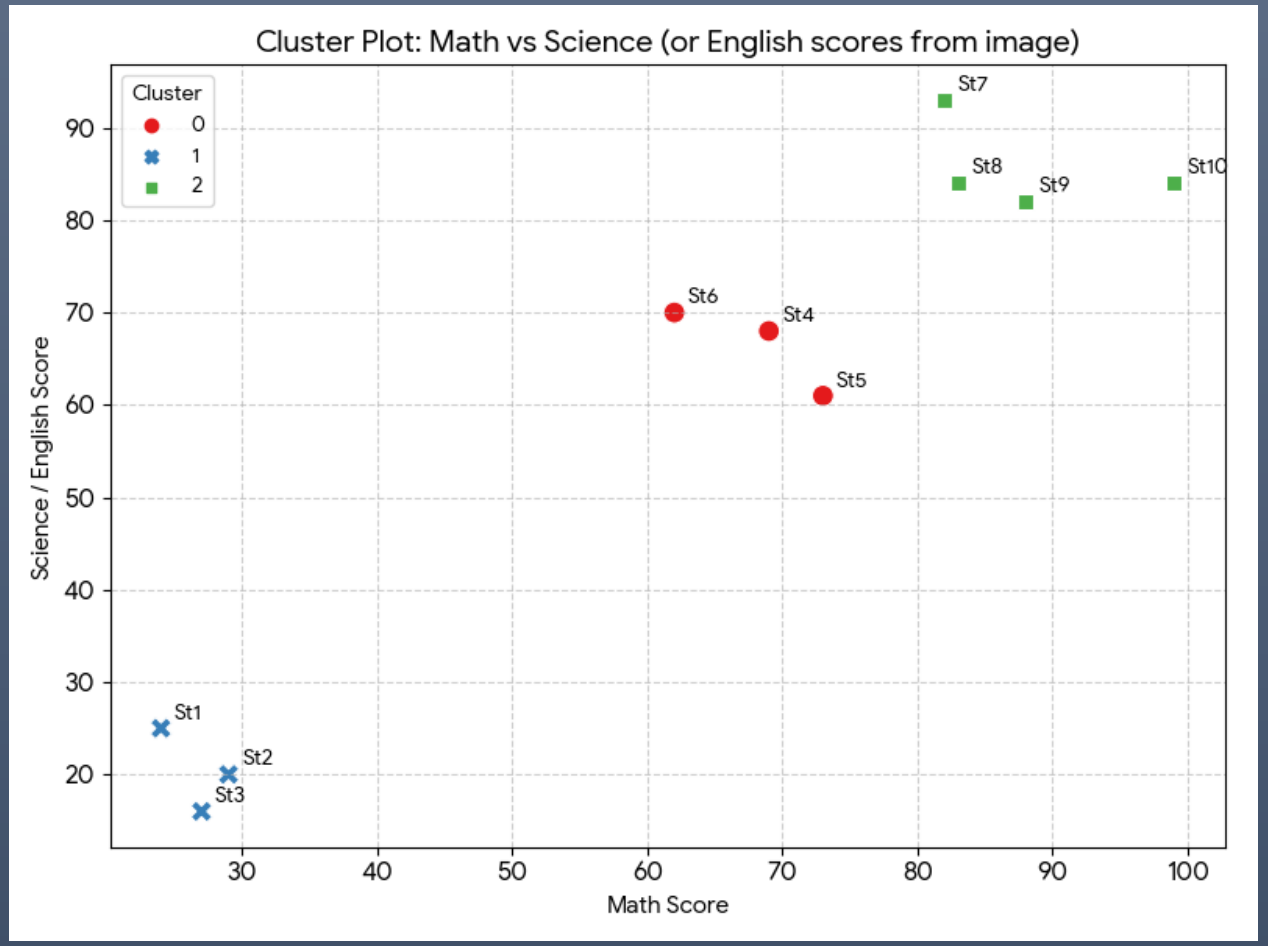
$$K = 3$$

$$C1 = (24, 25)$$

$$C2 = (69, 68)$$

$$C3 = (82, 93)$$

$$D = \sqrt{(x - c_x)^2 + (y - c_y)^2}$$



Student	Math	Science
St1	24	25
St2	29	20
St3	27	16
St4	69	68
St5	73	61
St6	62	70
St7	82	93
St8	83	84
St9	88	82
St10	99	84

K1			K2			K3	
Mean (Centroid Point)			Mean (Centroid Point)			Mean (Centroid Point)	
24	25		69	68		82	93

K1			K2			K3	
Mean (Centroid Point)			Mean (Centroid Point)			Mean (Centroid Point)	
26.5	22.5		69	68		82	93

$$K = 3$$

$$C1 = (24, 25)$$

$$C2 = (69, 68)$$

$$C3 = (82, 93)$$

$$D = \sqrt{(x - c_x)^2 + (y - c_y)^2}$$

Distance from C1:

$$D = \sqrt{(29 - 24)^2 + (20 - 25)^2}$$
$$= \sqrt{5^2 + (-5)^2}$$
$$= \sqrt{25 + 25}$$
$$= \sqrt{50}$$
$$= 7.07$$

Distance from C2:

$$D = \sqrt{(29 - 69)^2 + (20 - 68)^2}$$
$$= \sqrt{(-40)^2 + (-48)^2}$$
$$= \sqrt{1600 + 2304}$$
$$= \sqrt{3904}$$
$$= 62.48$$

Distance from C3:

$$D = \sqrt{(29 - 82)^2 + (20 - 93)^2}$$
$$= \sqrt{(-53)^2 + (-73)^2}$$
$$= \sqrt{2809 + 5329}$$
$$= \sqrt{8138}$$
$$= 90.21$$

Student	Math	Science
St1	24	25
St2	29	20
St3	27	16
St4	69	68
St5	73	61
St6	62	70
St7	82	93
St8	83	84
St9	88	82
St10	99	84

K1			K2			K3	
Mean (Centroid Point)			Mean (Centroid Point)			Mean (Centroid Point)	
26.7	20.3		69	68		82	93

$$D = \sqrt{(x - c_x)^2 + (y - c_y)^2}$$

Distance from C1

$$D = \sqrt{(27 - 24)^2 + (16 - 25)^2}$$
$$= \sqrt{3^2 + (-9)^2}$$
$$= \sqrt{9 + 81}$$
$$= \sqrt{90}$$
$$\boxed{D = 9.49}$$

Distance from C2

$$D = \sqrt{(27 - 69)^2 + (16 - 68)^2}$$
$$= \sqrt{(-42)^2 + (-52)^2}$$
$$= \sqrt{1764 + 2704}$$
$$= \sqrt{4468}$$
$$\boxed{D = 66.84}$$

Distance from C3

$$D = \sqrt{(27 - 82)^2 + (16 - 93)^2}$$
$$= \sqrt{(-55)^2 + (-77)^2}$$
$$= \sqrt{3025 + 5929}$$
$$= \sqrt{8954}$$
$$\boxed{D = 94.63}$$

EL-Bow:

WCSS-Within cluster sum of squared distance

